

College Management System (SDD)

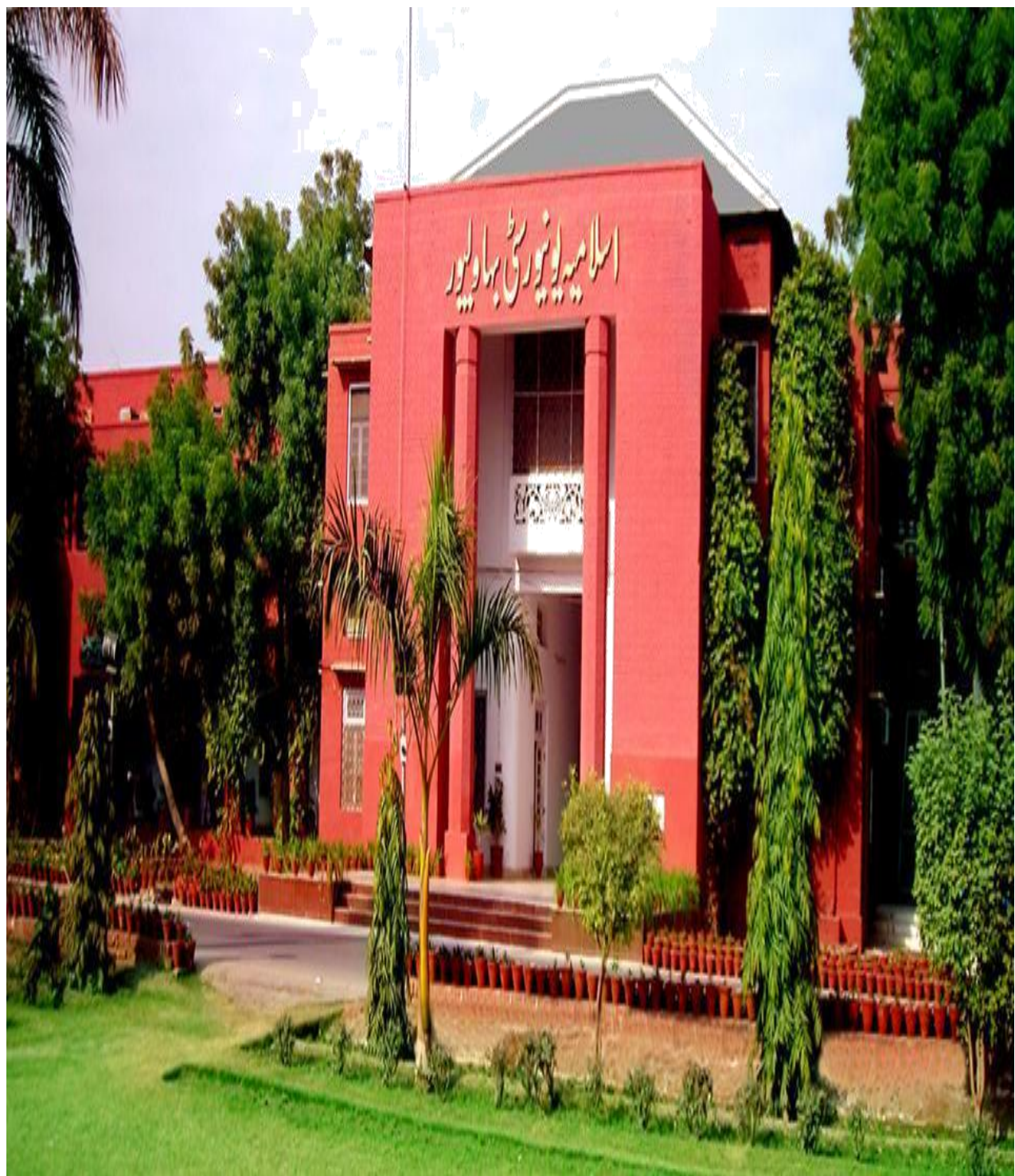
Software Design Document

Project Supervisor:	Sir Mirza Nasir
Developed by:	Sadia Shahbaz
Roll No:	23239555010
Session:	2023-2025



Department of Computer Science
Islamia University of
Bahawalpur





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Signature	Printed Name	Title	Date
	Sir Nasir Mirza	Supervisor, CSIT 21306	<date>

Table of Contents

CHAPTER NO 1: INTRODUCTION	7
1.1 Introduction	7
1.2 Purpose	7
1.3 Scope.....	8
1.4 Overview	8
1.5 Reference Material	8
1.6 Definitions and Acronyms.....	8
CHAPTER NO 2: SYSTEM OVERVIEW	10
2.1 Technologies Used	10
2.2 Application Overview.....	10
2.3 Design Languages.....	10
CHAPTER NO 3: SYSTEM ARCHITECTURE	11
3.1 Architectural Design.....	11
3.2 Decomposition Description.....	12
3.2.1Class Diagram:.....	12
3.2.2Activity Diagram:.....	13
3.3 Design Rationale	14
3.3.1 Collaboration Diagram:	14
3.3.2 Sequence Diagram:	15
CHAPTER NO 4: DATA DESIGN.....	16
4.1 Data Description	16
4.1.1 Zero Level Diagram	16
4.1.2 Context Level Diagram:.....	17
4.1.3 Detail Level Diagram	18
4.2 Data Dictionary	19
CHAPTER NO 5: COMPONENT DESIGN	32
5.1 COMPONENT DESIGN	32
CHAPTER NO 6: HUMAN INTERFACE DESIGN.....	33
6.1 Overview of User Interface	33
6.2 Screen Images	33
6.3 Screen Objects and Actions	37

CHAPTER NO 7: REQUIREMENT MATRIX	39
7.1 REQUIREMENTS MATRIX.....	39

CHAPTER NO 1: INTRODUCTION

1.1 Introduction

The title of this project is “**College Management System**” (CMS). CMS is Desktop application that aims at providing information to all the level of management within an organization. It monitors all aspects of college, its candidates, marks and other activities. It is simple application which allows the user to enter the data and also view its result. CMS is developed to provide the facility to the user to manage the record of the different teachers and students. In this, SDD shows how the software system will be structured to satisfy the requirement identifies in the software requirement specification. It is a translation into a description of the software structure, software component for the implementation phase. In essence, the SDD becomes a detailed blueprint for the implementation activity. In a complete SDD, each requirement must be traceable to one or more design entities.

1.2 Purpose

The main purpose of developing the “College Management System” (CMS) is to control the task related to the college candidate/employs. It is also used to minimize time to finding to require candidates in college record. “College Management System” (CMS) provides one attractive environment where you can manipulate data and information about candidate and staff easily. In this, the data is stored in an efficient way. Information about the students, teachers and other staff are easily kept in this system. All the record is permanently stored. Time saving and the correctness is the main benefit of this system.

The common features of the projects are:

- This is very easy to use for each user.
- Increase Efficiencies and Reduce Costs
- Transform IT for Higher Education
- Easy Solution
- Easy Admission Process
- Secure All Data
- Easy Account Maintenance
- Transaction History

1.3 Scope

“College Management System (CMS)” is used to search information of the user. This is used to login the system and to change the password after logging. This software is used to manage the student attendance and student time table. This software also manages the teacher’s record as well.

The requirement of the user is to:

- Interaction of system and operator.
- Admin will get access of checking buyer records, adds products and any modification after username & password authentication.
- Admin will add product modify update etc.
- Only admin have ability to access system.

1.4 Overview

This document will describe the design of various aspects of the project “including architectural of project, software components, interfaces, and database design. The 1st part of the SDD consist on the system functionality and interaction with administrator. The 2nd section defines the requirement specification. The “College Management System” is used to add the student and teacher record and manage the all data.

1.5 Reference Material

In this subsection:

- www.google.com.
- www.w3school.com.
- www.draz.pk.
- www.wikipedia.com
- www.Olx.com

1.6 Definitions and Acronyms

Admin	The system who is managing the system software.
--------------	---

User	The person who is using the system.
CMS	College Management System
SRS	Software Requirement Specification.
DBMS	Database Management System
C	C Sharp
GUI	Graphical user interface
SQL	Structured Query language

CHAPTER NO 2: SYSTEM OVERVIEW

2.1 Technologies Used

The system is coded with C# by using VISUAL STUDIO Tool. We will use local/internal database. This could be used in two ways: -

- Desktop application
- Window based application

We use Local/Internal database to store the data and record about the “College Management System”. “Desktop Application” could be used without internet and user with satisfied with application and useful for every user. “Window Based Application” could be used with internet and useful for online projects and online application.

2.2 Application Overview

The main objective of the project is to manage all the data and record of “College Management System”. This is consisting on Admin Login and User site. Admin Login used for Login to user and check user valid or not, admin can add and delete the data and then modify also. Admin and User both are participate with this application. User come from the college, visit this and then admin provide the objectives of the user.

- Admin must login for interact the system.
- When user login, then it receives the message valid or not.
- Admin can update any information.
- All record of user will save in database

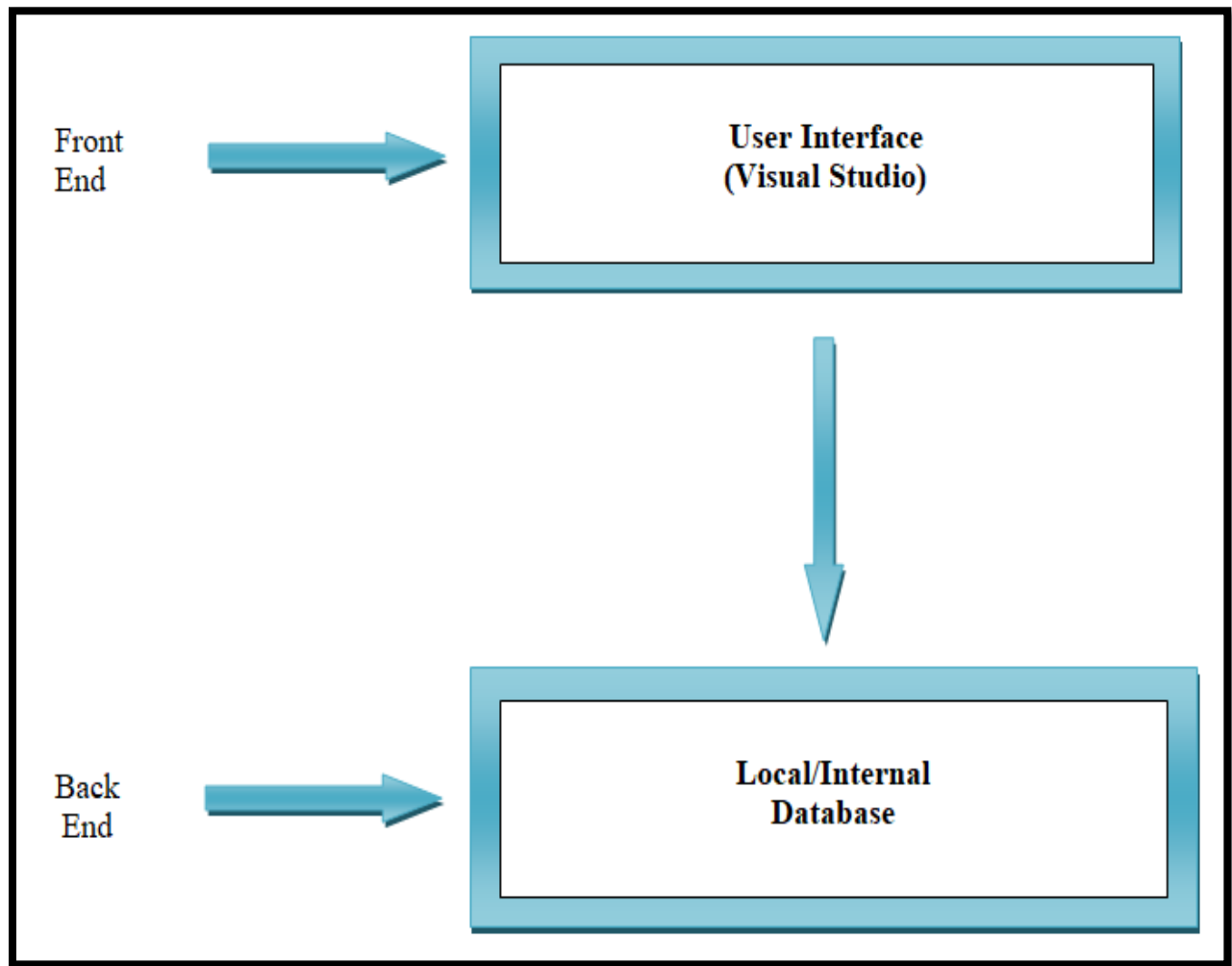
2.3 Design Languages

- C-Sharp(C#)
- Adobe Photoshop

CHAPTER NO 3: SYSTEM ARCHITECTURE

3.1 Architectural Design

The interface of system will design in C-sharp language by using visual studio. This application use for the College Management System for different purposes because it was developed in desktop application. This application is developed by using visual studio. All records of customers of students are used in internal/ offline database.

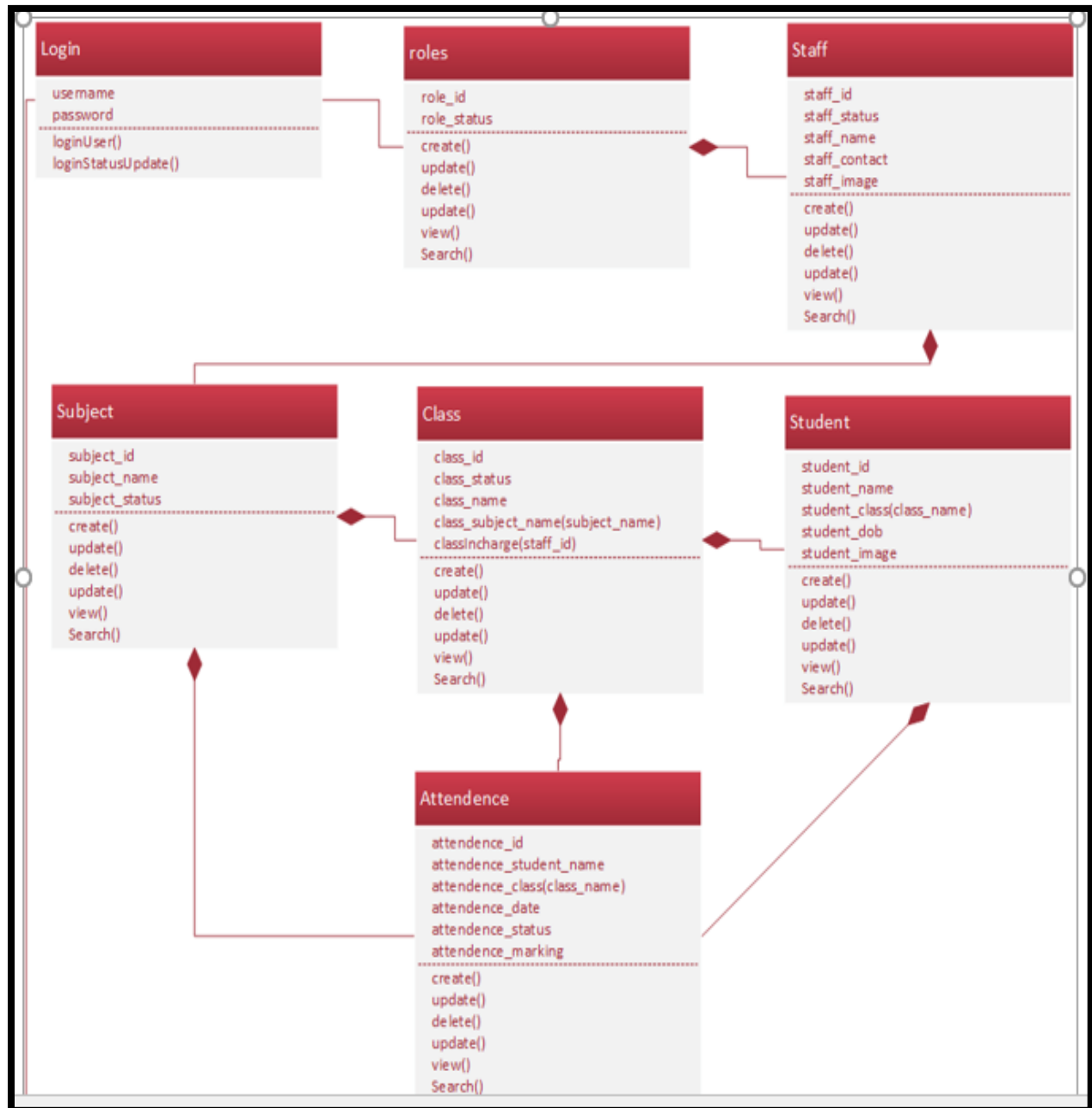


This diagram show front end and back end desktop application of College Management System.

3.2 Decomposition Description

In decomposition description top level data flow diagram (DFD) and structural decomposition diagrams.

3.2.1 Class Diagram:

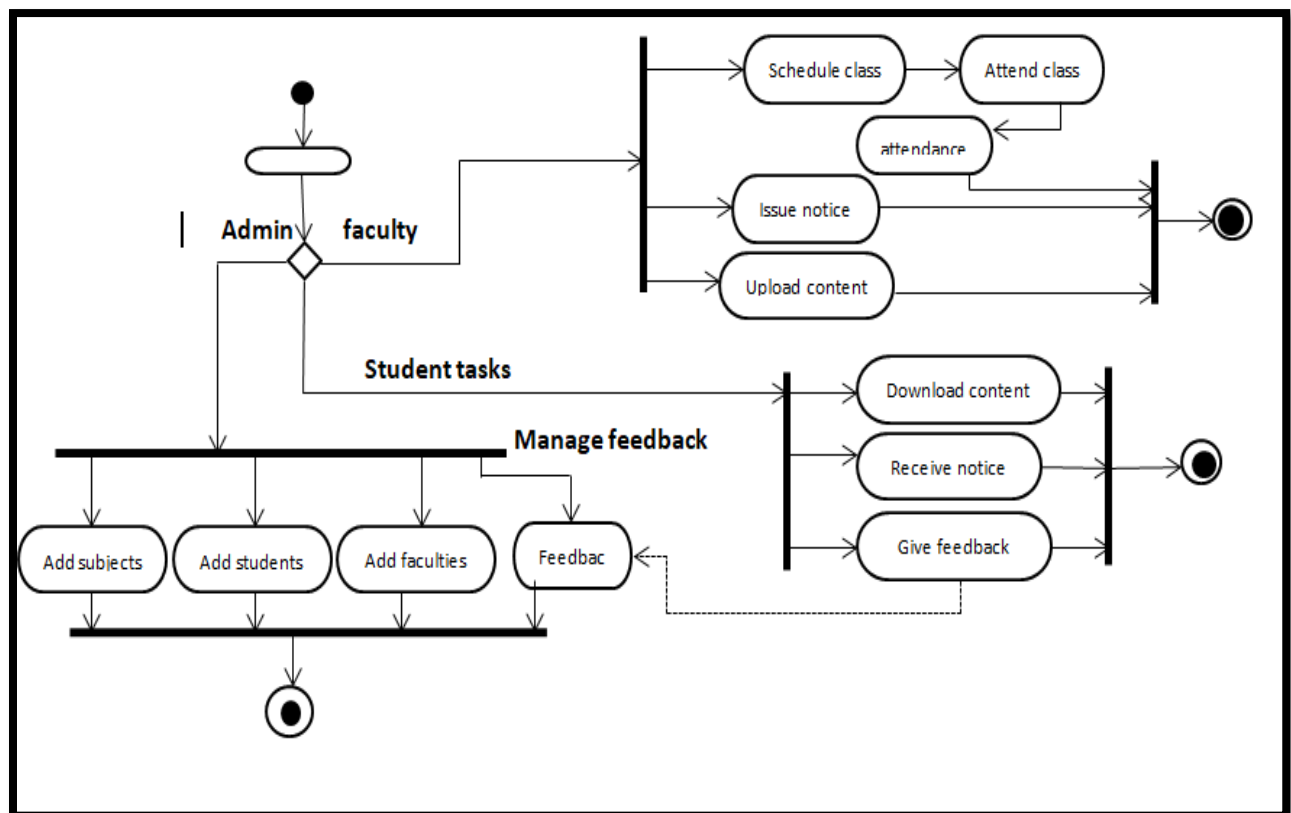


3.2.2 Activity Diagram:

Activity diagrams describe the workflow behavior of the system. These diagram are used in process modeling and analysis during requirements engineering. They are most useful for understanding work flow analysis of synchronous behaviors across a process. The diagrams describe the state of activities by showing the sequence of activities performed. They can show activities these are parallel or conditional. The main reason to use the activity diagram is to model the workflow behind the system being designed.

Components of Activity diagram:

Initial node	The filled in circle is the starting point of diagram. It does make it easier to read.
Final node	The filled circle with a border is the ending point of diagram. An activity diagram has zero or more ending point.
Flow	The flow is represented with the arrow in the diagram.
Fork	A fork is used when multiple activities are occurring at the same time.
Merge	A diamond with several flows entering and one leaving.
Join	A black bar with several flows and one entering.



3.3 Design Rationale

This application “College Management System” is a desktop application which will be developed/coded in C- sharp (C-sharp) language and visual studio is a tool for this language that is used for the application. This tool is very useful for this desktop application. To store or record to of data we use local /internal database by using. So saving of data activity is perform by local database.

On this request and assign spot to customer may reject to request due to incomplete data etc. If user request accepted, then assign the student No and store data in database of user record and in record of College Management System. Similarly, this operation can be performed by the admin and the admin can be modifying to data base.

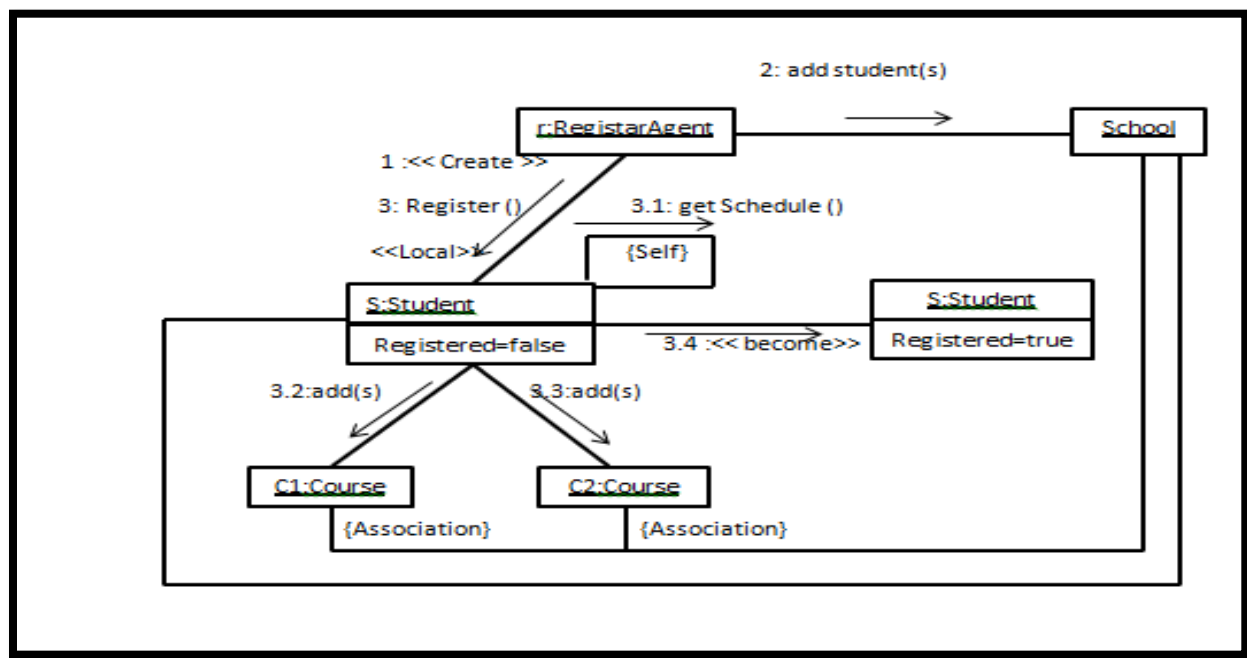
3.3.1 Collaboration Diagram:

Collaboration diagram illustrate interaction between objects. It also illustrates message being sent between classes and objects. These are very useful for visualizing the relationship between objects to collaboration to perform particular task. These provide good interaction between objects which may be difficult to see in class level.

Components of collaboration diagram: -

There are three primary component of collaboration diagram:

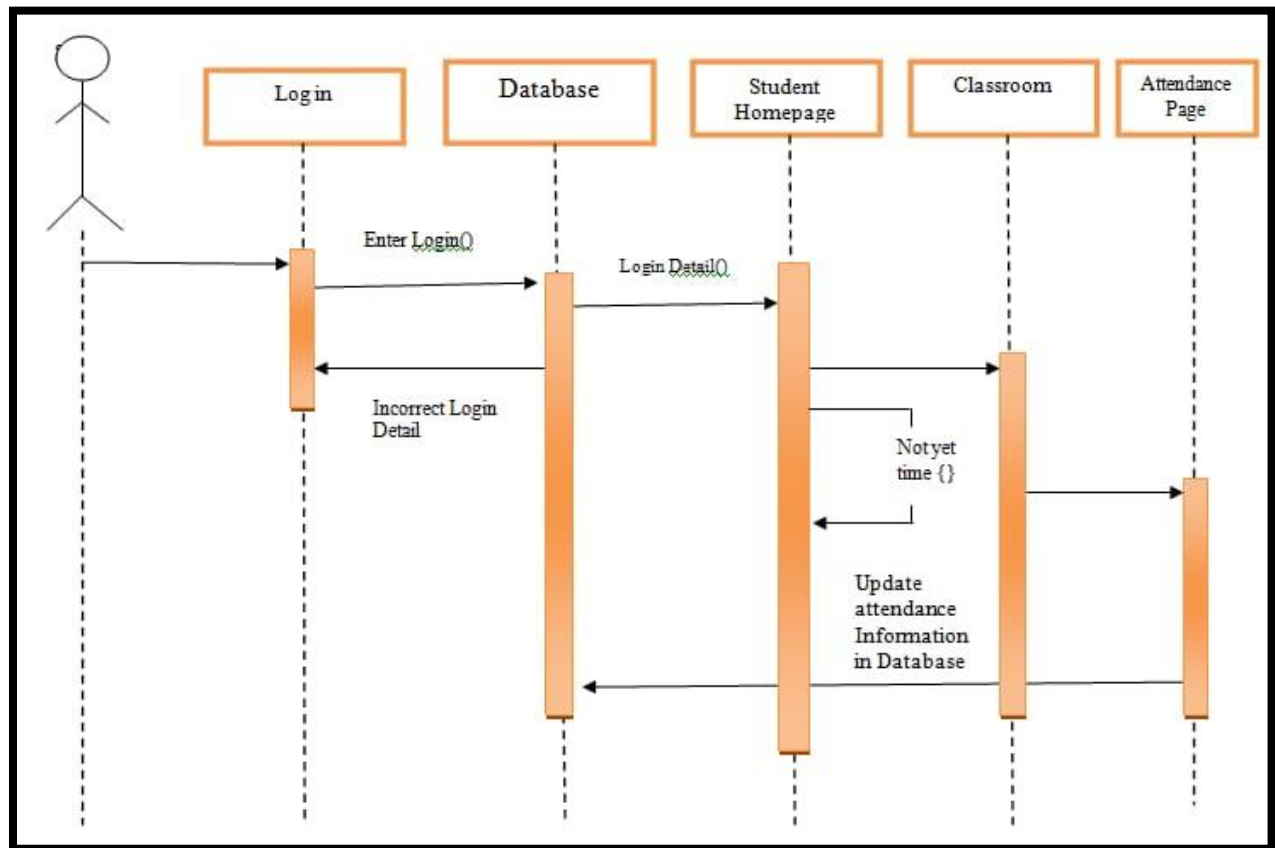
- Objects
- Link
- Message



3.3.2 Sequence Diagram:

As the name indicates, sequence diagram explain different sequence of different tasks. The purpose of this diagram is to show that the system is running in the proper way and nothing will go wrong at the time of executing.

AXIs	The x-axis shows the movement among different things and the y-axis shows the time.
COMMENT	Comment is used to explain what are going on the specific time or space. These are written on left side of the diagram.
INSTANCE	These are solid boxes on the top of the diagram with dotted lines.
CALL	These are the lines with the triangle at the end of each line. These are used to show movement of one instance to another.
RETURNS	These are used to show the return value of the function. Dotted line with an arrow on the edge is used to show the return.
GUARD STATEMENT	Guard statement is used to set some kind of criteria on any function.



CHAPTER NO 4: DATA DESIGN

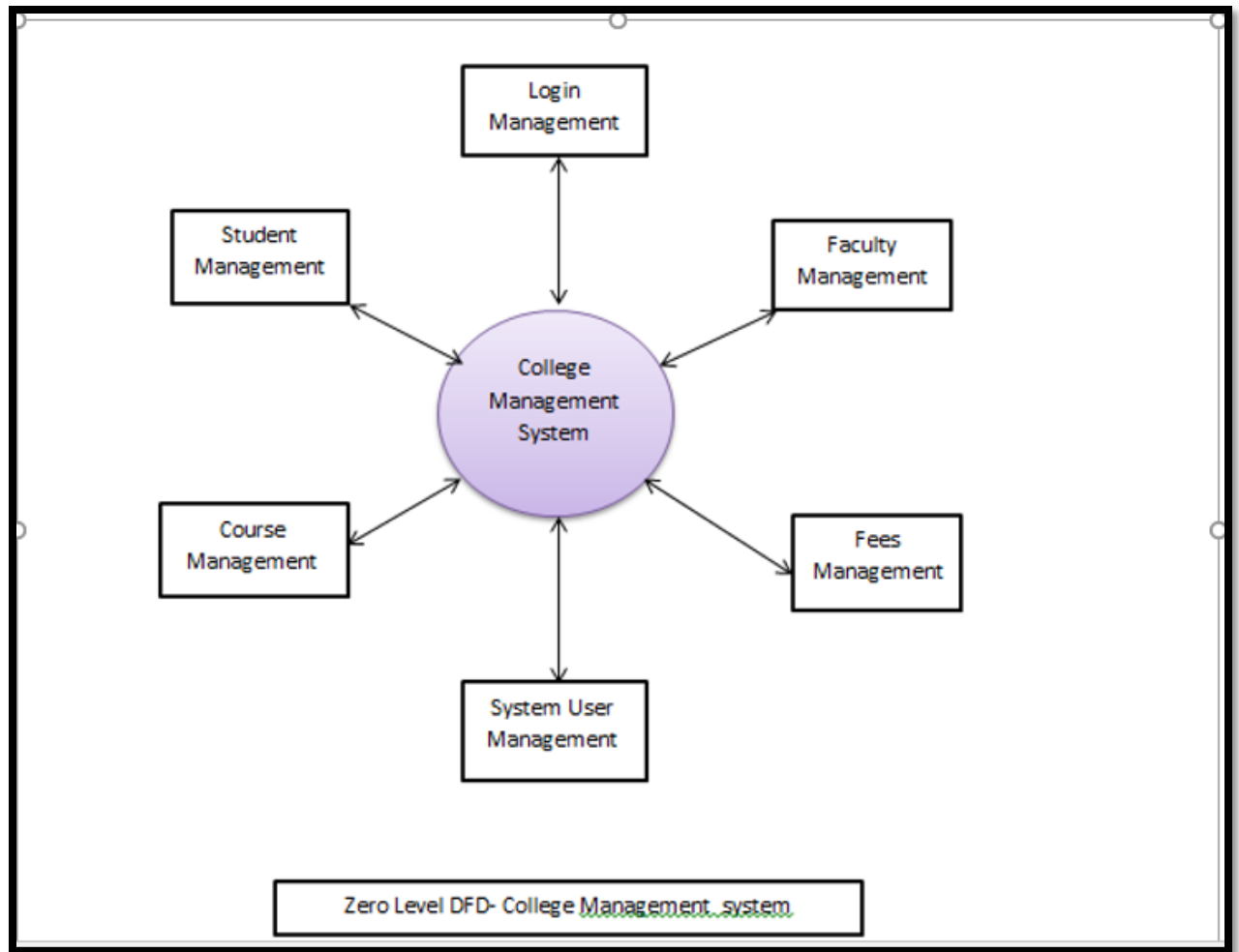
4.1 Data Description

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed and organized. List any databases or data storage items.

- Different constraints and checked are applied.
- Required data for students for students and teachers as conditions are stored in the database.
- We choose C-sharp language for developing and designing this application.
- VISUAL STDUIO tool is use for this application.

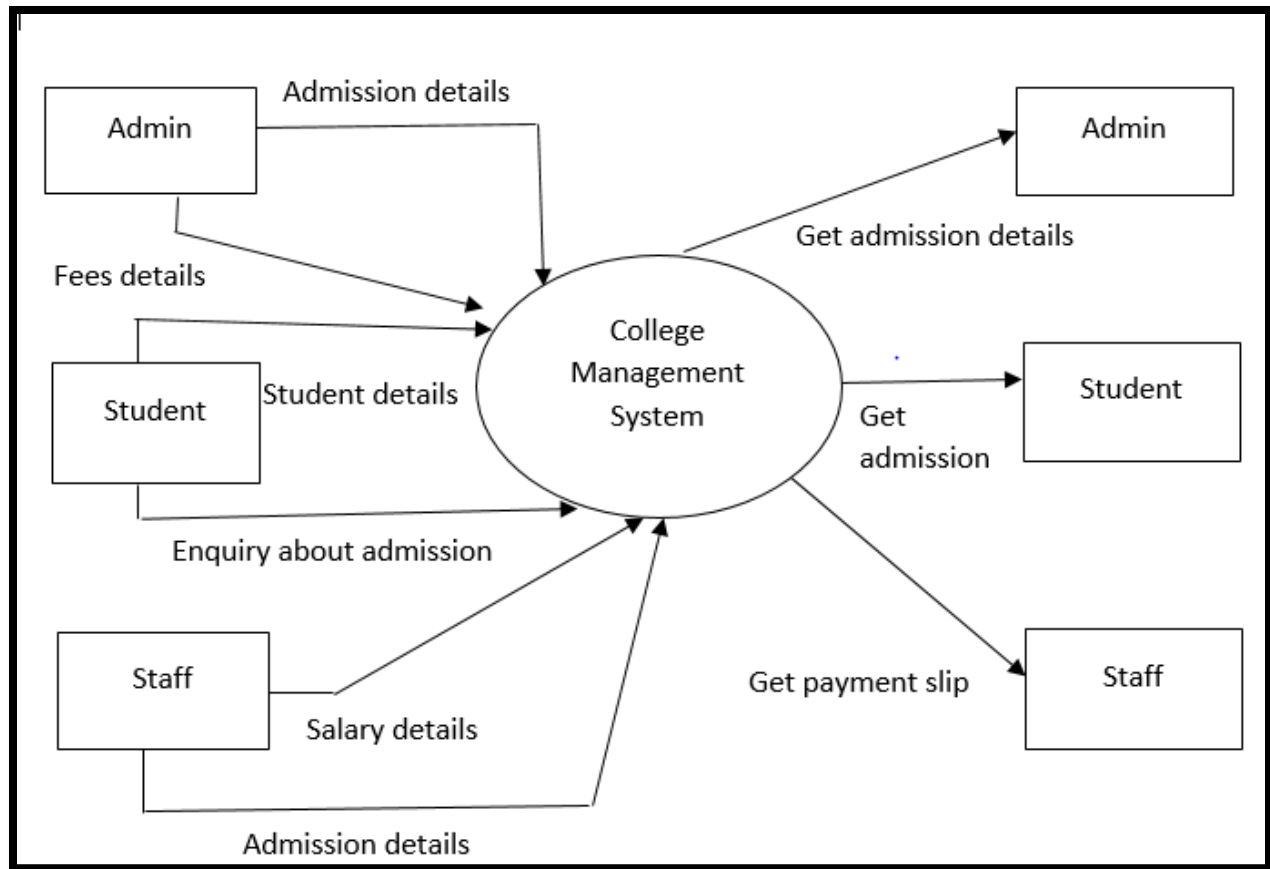
4.1.1 Zero Level Diagram

- (0-level) data flow diagram may only contain one process node (process-0) that generalizes the function of the entire system in relationship to external entities.
- A (0- level) diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.
- Level-0 context diagrams show a system as a single process with inputs and outputs flowing to or from external entities.

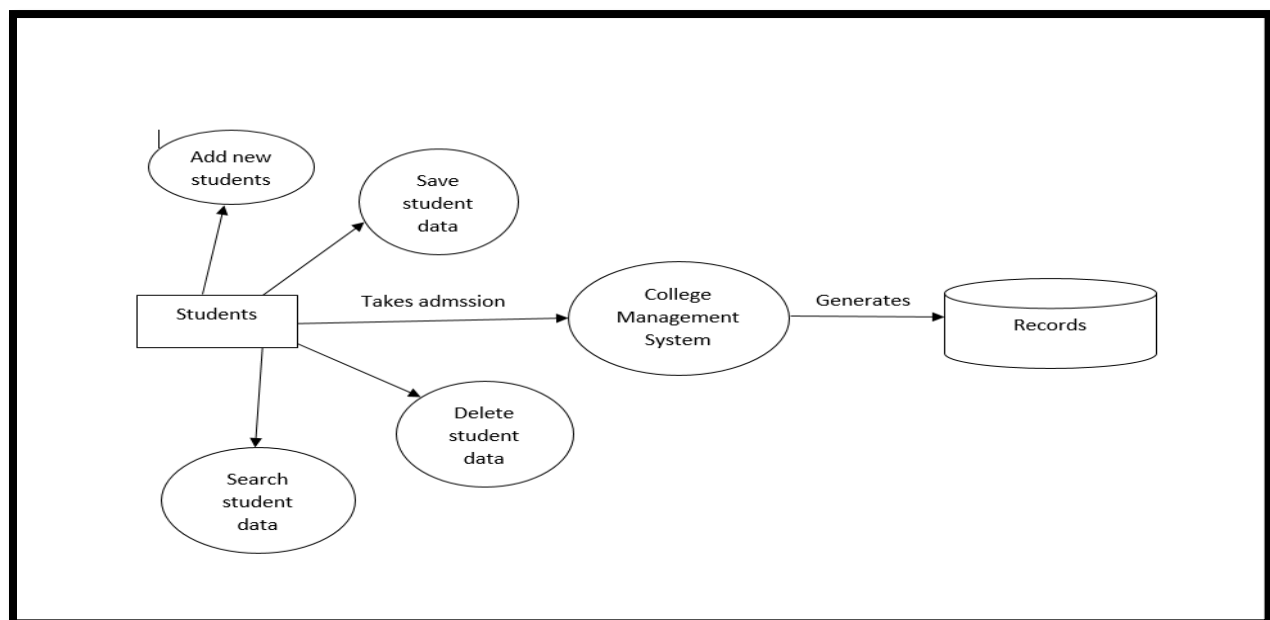


4.1.2 Context Level Diagram:

- Context level is a top level data flow diagram. It only contains one process node that Generalized the entire system.
- A context level diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.in external entities.
- It defines the boundary between the system and the part of the system.
- It exchanges data with outside entities. They are often used for high level planning.



4.1.3 Detail Level Diagram



4.2 Data Dictionary

Explanation of tables are as follows:

Data Dictionary of Admin Login Table:

- Username
- Username
- Username is used for storing the name of user
- nvarchar (50)
- 256 characters
- A to Z and a to z
- Password
- Password
- Password is used for storing the user password
- Admin-password
- int
- 0-9
- E-mail
- Email address
- Email is used for storing user email address
- nvarchar (50)
- 256 characters
- 0-9, a to z, A to Z, and special characters.

Table:

	Name	Data Type	Allow Nulls	Default	
	username	nvarchar(50)	☐		
	password	int	☐		
PK	Email	nvarchar(50)	☐		
			☐		

Query of this table:

```
CREATE TABLE [dbo].[AdminLogIn] (  
    [Username] NVARCHAR (50) NOT NULL,  
    [Password] INT NOT NULL,  
    [email] NVARCHAR (50) NOT NULL,  
    CONSTRAINT [PK_Table] PRIMARY KEY CLUSTERED ([email] ASC)  
);
```

Database of login table:

	Username	Password	email
PK	NULL	NULL	NULL

Data Dictionary of Add new students:

- Std Name
- Std Name
- It is used to store for student name
- nvarchar (50)
- 256 characters
- A to Z and a to z

- Std CNIC
- User CNIC
- It is used to store for student CNIC number
- Int
- 0 to 9

- E-mail
- E-mail Address
- It is used to store student email address
- nvarchar (50)
- 256
- A to Z, a to z, o to 9 and special characters

- Gender
- Std Gender
- It is used to identify student gender
- nvarchar (50)
- 256
- A to Z, a to z, o to 9

- Address
- Std Address
- It is used to store student address
- nvarchar (50)
- 256
- A to Z, a to z, o to 9

- Fees
- Std fees
- It is used for fees payment
- Int
- 0 to 9

- Std Date of Birth
- Std Date of Birth
- It is used for identify.
- Int
- 0 to 9

- Father Name
- Father Name
- It is used to store the father name
- nvarchar (50)
- 256
- A to Z, a to z, o to 9 and special characters

- Mobile
- Mobile
- It is used to add the contact
- Contact
- Int
- 0 to 9

Table:

	Name	Data Type	Allow Nulls
	std name	nvarchar(50)	☐
	std CNIC	int	☐
	Email	nvarchar(50)	☐
	Gender	nvarchar(50)	☐
	Address	nvarchar(50)	☐
	std date of birth	int	☐
	fees	nvarchar(MAX)	☐
	Father Name	nvarchar(50)	☐
	Mobile	int	☐
	Father CNIC	nvarchar(50)	☐
			☐

Query of this table:

```
CREATE TABLE [dbo].[ADD STUDENTS]
(
    [std name] NVARCHAR(50) NOT NULL,
    [std CNIC] INT NOT NULL,
    [Email] NVARCHAR(50) NOT NULL,
    [Gender] NVARCHAR(50) NOT NULL,
    [Address] NVARCHAR(50) NOT NULL,
    [std date of birth] INT NOT NULL,
    [fees] NVARCHAR(MAX) NOT NULL,
    [Father Name] NVARCHAR(50) NOT NULL,
    [Mobile] INT NOT NULL,
    [Father CNIC] NVARCHAR(50) NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY ([std CNIC])
)
```

Database of Add new students:

	std name	std CNIC	Email	Gender	Address	std date of birth	fees	Father Name	Mobile	Father CNIC
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Dictionary of Add new Teachers:

- Name
- Name
- It is used to store the teacher name
- nvarchar (50)
- 256 characters
- A to Z and a to z

- Mobile
- Mobile
- Contact
- Int
- 0 to 9

- E-mail
- E-mail Address
- Nvarchar(50)
- 256
- A to Z, a to z, o to 9 and special characters

- Gender
- Std Gender
- Nvarchar(50)
- 256
- A to Z, a to z, o to 9

- CNIC
 - User CNIC
 - Int
 - 0 to 9
-
- Father Name
 - Nvarchar(50)
 - 256
 - A to Z, a to z, o to 9 and special characters
-
- Address
 - Std Address
 - Nvarchar(50)
 - 256
 - A to Z, a to z, o to 9
-
- Qualification
 - Teacher qualification
 - Nvarchar(50)
 - A to Z, a to z and 0 9
-
- Salary
 - Teacher salary
 - Int
 - 0 to 9

Table:

	Name	Data Type	Allow Nulls	Default
	Name	nvarchar(50)	☐	
	Mobile	int	☐	
PK	CNIC	int	☐	
	Email	nvarchar(50)	☐	
	Gender	nvarchar(50)	☐	
	Address	nvarchar(50)	☐	
	Qualification	nvarchar(50)	☐	
	Fathername	nvarchar(50)	☐	
	Salary	int	☐	

Query of this Table:

```
CREATE TABLE [dbo].[Table] (
    [Name] NVARCHAR (50) NOT NULL,
    [Mobile] INT NOT NULL,
    [CNIC] INT NOT NULL,
    [Email] NVARCHAR (50) NOT NULL,
    [Gender] NVARCHAR (50) NOT NULL,
    [Address] NVARCHAR (50) NOT NULL,
    [Qualification] NVARCHAR (50) NOT NULL,
    [Fathername] NVARCHAR (50) NOT NULL,
    [Salary] INT NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY CLUSTERED ([CNIC] ASC)
);
```

Database of Add new table:

Name	Mobile	CNIC	Email	Gender	Address	Qualification	Fathername	Salary
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Dictionary of Students Attendance table:

- Std Name
- Std Name
- It is used to store the student name
- nvarchar (50)
- 256 characters
- A to Z and a to z

- Roll no
- Std Roll no
- It is used to store the student roll no
- Int
- 0 to 9

- Campus Time
- Time
- It is used to set the time
- Int
- 0 to 9

- Fine
- Std Fine
- It is used to store the fine
- Int
- 0 to 9

- Date

- Date
- It is used to set the date
- int
- 0 to 9

Table:

	Name	Data Type	Allow Nulls
	Std Name	nvarchar(50)	☐
	Rollno	int	☐
	Campous time	int	☐
	Fine	int	☐
	Date	int	☐

Query of this table:

```
CREATE TABLE [dbo].[std_Attendance]
(
    [Std Name] NVARCHAR(50) NOT NULL,
    [Rollno] INT NOT NULL,
    [Campous time] INT NOT NULL,
    [Fine] INT NOT NULL,
    [Date] INT NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY ([Rollno])
)
```

Database of Add new Teachers table:

	Std Name	Rollno	Campous time	Fine	Date
*	NULL	NULL	NULL	NULL	NULL

Dictionary of Fees table:

- Std Name
- Std Name
- nvarchar (50)
- 256 characters
- A to Z and a to z
- Total Fees
- Fees
- It is used to store the fees
- Int
- 0 to 9
- Paid Fees
- Fees
- It is used to store the paid fees
- Int
- 0 to 9
- Unpaid Fees
- Fees
- It is used to store the unpaid fees
- Int
- 0 to

Table:

	Name	Data Type	Allow Nulls	Default
1	Std name	nvarchar(50)	☐	
2	Total fees	int	☐	
3	Paid Fees	int	☐	
4	Unpaid	int	☐	
5			☐	

Query of this table:

```
CREATE TABLE [dbo].[Fees] (
    [Std name] NVARCHAR (50) NOT NULL,
    [Total fees] INT NOT NULL,
    [Paid Fees] INT NOT NULL,
    [Unpaid ] INT NOT NULL,
    PRIMARY KEY CLUSTERED ([Std name] ASC)
);
```

Database of Fees table:

	Std name	Total fees	Paid Fees	Unpaid
*	NULL	NULL	NULL	NULL

4.3 E-R Diagram:

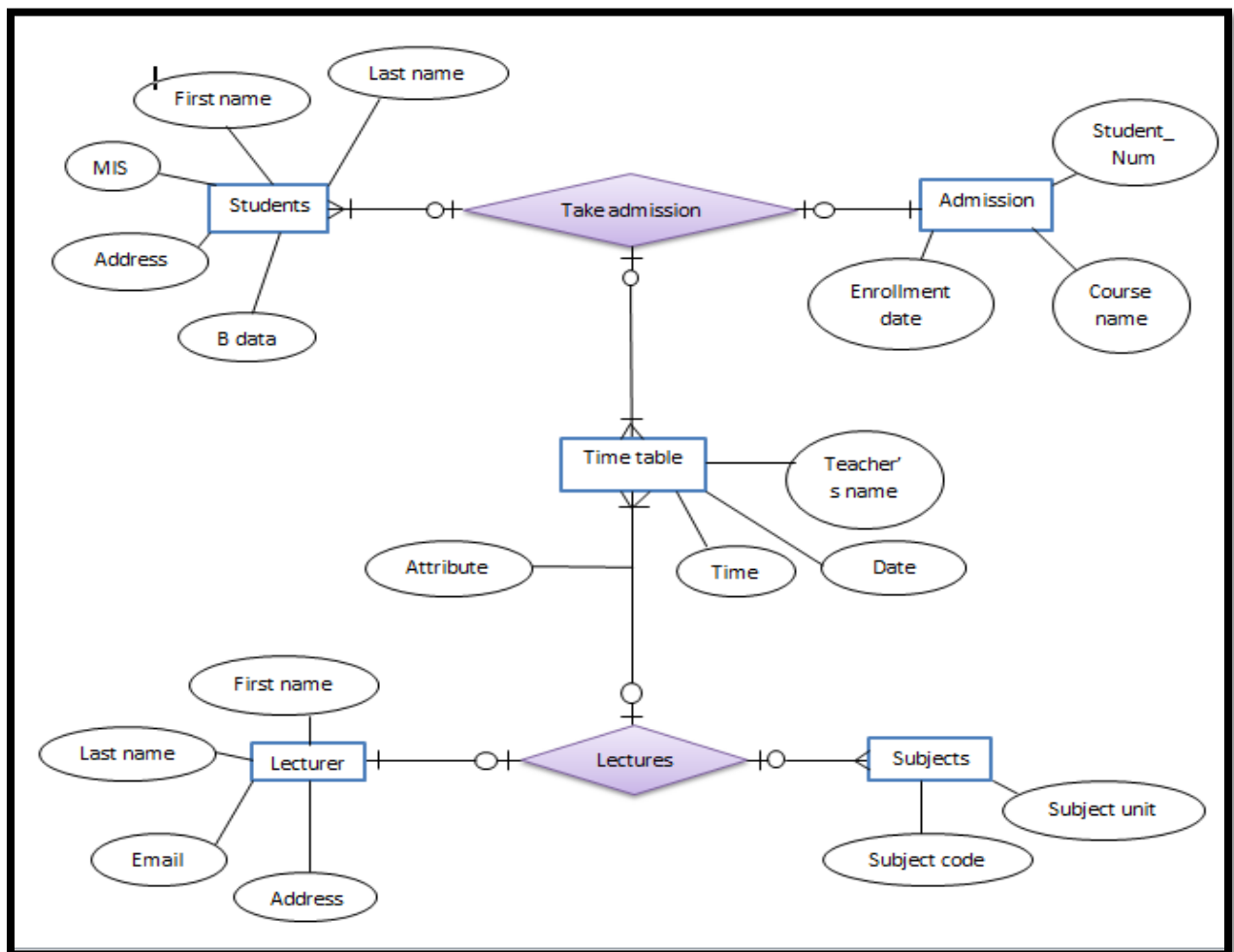
ER stands for “Entity Relationship diagram.” An E-R diagram efficiently shows the relationship between various entities stored in a database. E-R Diagram are used to model real world objects like a person, a car, a company etc. It has some features:-

- E-R diagram are used to represent the E-R model in a database, which makes them easy to convert into relationship.
- E-R diagram provide purpose of real world modeling of objects which makes them intently useful.
- E-R diagrams require no technical knowledge & no hardware support.

- It gives a slandered solution of visualizing the data logically.
- These diagrams are very easy to understand.

E-R diagram symbols and annotation:-

	Entity	Any real world object can be represented as an entity which data can be stored in a database.
	Relationship	Any entity has a set of properties. These properties of each entity are termed as attribute.
	Attribute	A relationship is defined as bond or attachment between 2 or more entities.



CHAPTER NO 5: COMPONENT DESIGN

5.1 COMPONENT DESIGN

- The system is capable to do work according to the direction.
- Every simplest or complex component is responsible to do work which is assigned to it.
- The system considers every component as best efficient way component for the mutual cooperation of the whole
- Process. Component design follows as:-



The short detail of the component design working describes as the admin after login to the system will choose an option for the submission or other action. Consider an example if admin want to generate bill then admin will press the button of total bill. The given data in the text box will be stored in the database, similarly admin can do updating by the corresponding button in it.

CHAPTER NO 6: HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

It describe the functionality of the system from the user's perspective. It also explain how the user will be able to use your system to complete all the expected features and the feedback information that will be displayed for the user.

6.2 Screen Images



The image shows a graphical user interface for a login system. The window has a title bar with the text 'LOGIN' and standard window controls (minimize, maximize, close). The main area contains two text input fields, one labeled 'Enter Name' and the other 'Password'. Below the input fields are two large circular buttons. The left button features a blue padlock icon, and the right button features a yellow open padlock icon. At the bottom center of the window is a rectangular button labeled 'Login'.

Login Form



Main Form:

The screenshot shows a window titled "Add Student" with standard Windows window controls. The main content area has a blue background with a floral pattern and is titled "Student Admission". It contains several input fields and buttons arranged in two columns. The left column includes "Std Name", "Std CNIC", "Email", "Gender" (with radio buttons for "Male" and "Female"), "Address", and "Fees". The right column includes "Date of Birth", "Father Name", "Father CNIC", and "Mobile". At the bottom right, there are "Save" and "Reset" buttons. Close buttons (X) are present in the top-left and bottom-right corners of the form area.

Student Admission Form

Attendance

Std Attendance

Std Name

Class

Rollno

Campous Time

Fine

Date

Save

Reset

Student Attendance Form

Fees

Fees

Std Name

Total Fees

Paid Fees

Unpaid

Save

Reset

Student Fees Form

New_Teacher

Add New Teacher

Name **Qualification**

Mobile **Address**

Email **Salary**

Gender ☐ Male ☐ Female

Fathername

CNIC

Save **Reset**

Add New Teacher

LOGOUT

Logout

Logout

Logout Form:

6.3 Screen Objects and Actions

A discussion of screen objects and actions associated with those objects.

Login Form:

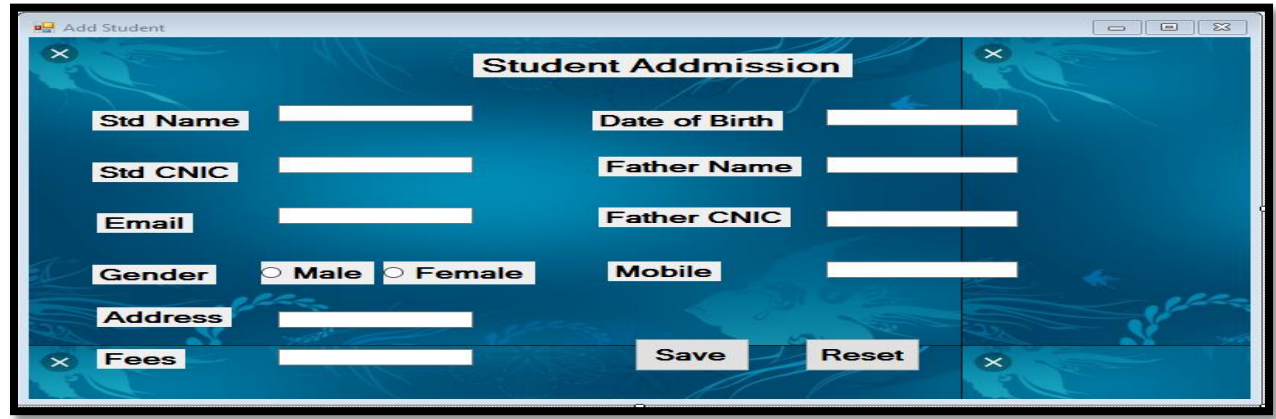
Login form is used for admin. It will be used for every time when system will shut down and restart this application. This form is necessary for interact with system.



Login Form

Student Admission:

This form is fill the students will take the admission. This is very important to fill for the student. It will provide the important information about students.

The image shows a software window titled 'Add Student' with a blue background and a decorative pattern. The window contains a form titled 'Student Admission'. The form has two columns of input fields. The left column includes 'Std Name', 'Std CNIC', 'Email', 'Gender' (with radio buttons for 'Male' and 'Female'), 'Address', and 'Fees'. The right column includes 'Date of Birth', 'Father Name', 'Father CNIC', 'Mobile', and 'Save' and 'Reset' buttons. There are 'X' buttons in the top-left and top-right corners of the form area.

Student Admission Form

Add new teacher:

This form is for the teacher to apply for job. In this, complete information are provide through this. This is the very important form for teacher.

The image shows a software window titled 'New_Teacher' with a background image of a teacher in a classroom. The window contains a form titled 'Add New Teacher'. The form has two columns of input fields. The left column includes 'Name', 'Mobile', 'Email', 'Gender' (with radio buttons for 'Male' and 'Female'), 'Fathername', and 'CNIC'. The right column includes 'Qualification', 'Address', 'Salary', and 'Save' and 'Reset' buttons. There are 'X' buttons in the top-left and top-right corners of the form area.

Teacher Form

CHAPTER NO 7: REQUIREMENT MATRIX

7.1 REQUIREMENTS MATRIX

It provide a cross reference that traces components and data structures to the requirements in your SDD document. Use a tabular format to show which system components satisfy each of the functional requirements from the SDD. Refer to the functional requirements by the numbers/codes that you gave them in the SDD.

Components: Requirements from SRS (Use case):	Login logout Component	Main Functional ity Component	Settings Component	Social Component	Trade Component	Data Services Component	Remote Server Component
UC1	X	X					
UC2	X	X					
UC3	X	X					
UC4	X	X	X				
UC5	X		X				X
UC6		X					
UC7	X	X					